

Programmer's Model of 8086

Q. Describe the programmer's model of the 8086 microprocessor. Detail the General Purpose Registers, Segment Registers, Pointer/Index Registers, and the Flag Register.

****Definition to Remember****

The ****P**
divided

1. General Purpose Registers (16-bit / 8-bit split)

Can be used as one 16-bit register or two 8-bit registers (High/Low).

* ****AX**

2. Segment Registers (16-bit)

Hold the base addresses of 64 KB memory segments to help address 1 MB of physical memory.

* ****CS**

3. Pointer and Index Registers (16-bit)

Hold offset addresses relative to the segment registers.

* ****SP**

4. Flag Register (16-bit)

Contains 9 active flags (6 Status, 3 Control).

* ****Sta**

****Must-Write Points (for 10 marks)****

1. Conta

Quick Recall

Model
CF/PF/A